VAATSALYA MISHRA

Computer Science (AI) Undergraduate | Software Engineering (SWE) • Data Analytics • AI & Cybersecurity

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PROFESSIONAL SUMMARY

Third-year B.Tech student in Computer Science & Engineering (Artificial Intelligence) at PSIT Kanpur with a strong foundation in Data Structures & Algorithms (DSA), problem-solving in Python and C++, Machine Learning fundamentals, and Information Security. Currently working as a Data Analyst Intern at Bluestock Fintech cleaning datasets, writing SQL queries, and building BI dashboards (Power BI, Tableau, Excel), while engineering independent multi-agent AI systems and high-efficiency algorithmic tools. Seeking Software Engineering (SWE), Data Analytics, and AI/ML opportunities.

TECHNICAL SKILLS

Programming & Development: Python, C++, SQL, JSON, HTML/CSS; *Currently Learning:* JavaScript, React

Core CS, AI & Security: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Artificial Intelligence (AI), Machine Learning Fundamentals, Information Security, Cybersecurity

Data Analytics & BI: Python (pandas, NumPy), Exploratory Data Analysis (EDA), Data Cleaning & Validation, Data Interpretation, KPI Reporting, Data Visualization, Power BI, Tableau, Excel, Google Sheets

Cloud & Developer Tools: Oracle Cloud Infrastructure (OCI), Git, GitHub, Ollama

EXPERIENCE & INDUSTRY SIMULATIONS

Data Analyst Intern — Bluestock Fintech
Sep 2026 – Nov 2026

- Clean, validate, and structure live fintech datasets and write optimized SQL queries to pull accurate data for recurring analytics and cross-functional reporting.
- Build interactive charts and dashboards using Power BI, Tableau, and Excel, authoring analytical summaries to surface key financial trends and anomalies.

Virtual Job Simulations — Deloitte Australia (Data Analytics) & TCS × Forage (Cybersecurity)
2026

- Completed industry job simulations in forensic data analysis, client dashboarding, and enterprise cybersecurity threat assessment across Deloitte Australia and TCS workflows.

SOFTWARE ENGINEERING & AI PROJECTS

Self-Learning AI Agent | *Python, Ollama, JSON*
Personal Project

- Architected a multi-agent orchestration system running locally with zero API or cloud cost, routing research tasks to one sub-agent and coding tasks to another.
- Designed a persistent JSON memory layer that extracts conversational facts and stores structured context across sessions without a vector database.
- Engineered a deterministic JSON function-calling loop from scratch enabling lightweight open-source models to reliably execute web search, code execution, and file reading tools.

Smart Task Scheduler | *Python, heapq, Hash Map, Stack, JSON* **Personal Project**

- Built a CLI task scheduling engine ordering tasks by priority and deadline using a binary min-heap (`heapq`) for $O(\log n)$ retrieval performance.
- Implemented hash map indexing for $O(1)$ lookups and a stack-based undo mechanism with automated JSON persistence across terminal sessions.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering (Artificial Intelligence) — PSIT Kanpur **Sep 2024 – Sep 2028**

CERTIFICATIONS & HACKATHONS

- **AI & Cloud Certifications:** Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate • AWS Educate: Introduction to Generative AI • YUVA AI For All (TCS iON × IndiaAI, Sep 2026)
- **Data & Programming Certifications:** Exploratory Data Analysis (Infosys Springboard, Sep 2026) • Python Foundation Certification (Basics of Python) • Practical HTML & CSS Assessment (81.25%, Infosys Springboard)
- **Hackathons & Coding Assessments:** Adobe University Hackathon 2026 (Round 1 MCQ & Coding) • TVS Credit E.P.I.C 8.0 IT Challenge (Round 1 Assessment) • National Online Hackathon 2026 (Startup) • Idea2Impact 2026